Java LAB Practice Programs based on1- 5 Units (Java Practical File)

Prepare file in Word Document (Question, Code, Output Screenshot)

1) WAP in Java to sum of 2 no in diff ways (3-simple in main, using method, using constructor).

2) WAP in Java to elaborate different method of taking input to find the sun of 2 nos. (Scanner, Buffered Reader, Command Line arguments).

3) WAP in Java to calculate area of a rectangle using (Class, object, Demonstrate different methods of initialising an object)

4)WAP to calculate area and volume of a box use all the types of constructor (Demonstrate constructor overloading).

5) WAP in Java to perform the simple inheritance. (Define a class Student, with basic details- RollNo, Name, Course . Define supportive constructors and methods. Define the class score that inherits class student and calculate average marks in 5 subjects, and calculate grade)

6) WAP in Java to perform the multilevel inheritance. (Define class Vehicle ( )- no_Wheels, FeulType, define subclass Car - Company, Define subclass SUV- Model, Capacity. Price) Define supportive Constructors & methods.

7) Create a class customer (cust I'd, name, address, phn no) and sub class Bank account (type of account, ac no, balance, deposit (), withdraw ()).

8) WAP to demonstrate the uses of super keyword. [WAP to define a class Score, Calculating avg marks of 3 subject of a student. Use constructor in both classes]

9) WAP to demonstrate multiple uses of This keyword. Create a class Student with instance variables name and age.

Provide:

- A default constructor.
- A parameterized constructor that calls the default one using this().
- A method display() to print student details.
- Create a method getStudent() that returns the current object using this.
- Create a method updateAge(int age) that updates the age and returns the object using this (method chaining).
- Create another class Helper with a method printDetails(Student s) to display details of a passed Student object.

Demonstrate all these uses in the main() method.

Q10. Write a Java program to demonstrate the concepts of interfaces:

Requirements

- Create an interface Vehicle with the following:
  - A constant MAX_SPEED.
  - Abstract methods start(), stop(), and getDetails().

- Create another interface Maintenance with a method service().
- Create a class Car that:
  - Implements both Vehicle and Maintenance interfaces.
  - Defines attributes like model and year.
  - Provides concrete implementations for all abstract methods.
- Create another class Bike that:
  - Implements only the Vehicle interface.
- In the main() method:
  - Create objects of Car and Bike.
  - Use Vehicle interface reference to call methods.
  - Demonstrate accessing MAX_SPEED.
  - Show polymorphism by storing Car and Bike objects in a Vehicle[] array and calling their methods dynamically.

Q 11) WAP in Java to perform boxing in all primitive data type .

Q 12) WAP in Java to perform Unboxing using all wrapper class .

Q13 write a Java program to demonstrate defining and accessing Java packages

Q 14.) WAP in Java to demonstrate the concept of array list using diff methods and perform following operation.

1)To insert an element

2)At the particular index

3) To retrieve at an index

4) To change the value at q specific index

5) to delete an element at the end

6) delete all the elements

7) to iterate the elements for using for loop

8)to iterate the elements for each loop

9) create a clone of array list

10) To sort an array list

Q15.) WAP in Java to create an array list storing the character values and print all the vowels from the list

Q 16) WAP in Java to demonstrate the concept of LinkedList collection

using diff methods and perform following operation.

1)To insert an element

2)At the particular index

3) To retrieve at an index

4) To change the value at q specific index

5) to delete an element at the end

6) delete all the elements

7) to iterate the elements for using for loop

8)to iterate the elements for each loop

9) create a clone of LinkedList

10) To sort a LinkedList

Q17 Create a ArrayList storing integers and print all the prime numbers.

Q.18) WAP to create a phone directory storing the name of a customer as a key and contact no as a value (both are string type) and perform the following operation (menu driven)

1) insert an element.

2)To retrieve the element.

3) Display all the keys (using keyset)

4) Display all the keys and values .

5)To delete an element.

6) Iterate using for each loop .

7)To search an element in the HashMap(take input from the user).

8) perform Sorting in the HashMap.

9)Clear all the elements.

Q .19) Design a **menu-driven program** in Java to manage **student records** using **TreeMap**.

Requirements:

1. **Data Structure:**
   - Each student record contains:
     - **Roll Number (int)** → Key in TreeMap
     - **Name (String)** → Value
     - **Marks (float)** → Can be part of a Student object as the value
2. **Menu Options:**

Option	Description
1	**Add Student Record** – Add a student with roll number, name, and marks.
2	**Delete Student Record** – Remove a student using roll number.
3	**Search Student Record** – Search by roll number and display student details.
4	**Display All Records** – Display all students sorted by roll number.

5 **Display Students in Range** – Display students with roll numbers between x and y.

6 **Average Marks** – Calculate and display average marks of all students.

0 Exit

3. **Implementation Details:**
    - Use **TreeMap<Integer, Student>** where Student is a class containing name and marks.
    - TreeMap ensures **records are automatically sorted by roll number**.
    - Duplicate roll numbers should **update existing record**.
    - Use iterators or subMap() for **range-based display**.
    - Program should **loop until the user chooses to exit**.

Q20. Write a Java program that performs the following tasks:

1. Create a HashSet of **student names** and add at least 6 names, including duplicates.
2. Display all elements of the HashSet.
3. Check whether a specific student (entered by the user) is in the HashSet.
4. Remove a student name from the HashSet (entered by the user).
5. Create a TreeSet of **student marks** and add at least 7 marks, including duplicates.
6. Display all elements of the TreeSet (marks should appear in **sorted order**).
7. Display the **highest** and **lowest** marks from the TreeSet.

Q21 .Practical Question: Multithreading in Java

Write a Java program to demonstrate **multithreading** by performing the following tasks:

1. Create **two threads**:
    - **Thread 1**: Prints numbers from **1 to 10** with a **1-second delay** between each number.
    - **Thread 2**: Prints letters from **A to J** with a **1.5-second delay** between each letter.
2. Both threads should run **simultaneously**, i.e., numbers and letters should interleave in the console output.
3. Use either:
    - **Extending Thread class**, or
    - **Implementing Runnable interface**

Q22. Practical Question: Multithreading in Java (Comprehensive)

1. Write a Java program to demonstrate **multithreading** using the following methods:
   a) **Extending the Thread class**
   b) **Implementing the Runnable interface**
   c) Using **wait() and notify()** for synchronization
   d) Using **deprecated methods suspend() and resume()** (for educational purposes only)
2. In each case, implement a program that:
    - Prints **even numbers from 2 to 20** in one thread.
    - Prints **odd numbers from 1 to 19** in another thread.
    - Ensure that **threads run concurrently** and numbers are **interleaved**.

3. For wait()/notify(), ensure **alternate printing** of even and odd numbers.
4. For suspend()/resume(), ensure threads **alternate safely** (acknowledge the deprecation warning).
5. Submit **four separate programs** demonstrating each method with **output**.

Q23 Create a Java program to demonstrate thread synchronization.

• Create a class PrintTable with a synchronized method print(int n) that prints the multiplication table of n.

• Create two threads that share the same PrintTable object and call the print() method with different values.

Q24 Program to create a Java Swing application that displays buttons for different cursor types and changes the cursor shape dynamically when each button is clicked.

Q 25 Design a GUI-based calculator using Java Swing that can perform addition, subtraction, multiplication, and division of two numbers.

Q 26. Student Registration Form
Create a Swing-based registration form that takes Name, Roll Number, Gender, and Course, and displays the entered data on submit.

Q27
a) Demonstrates applet lifecycle methods (init(), start(), stop(), destroy(), paint())
b) Uses mouse events to draw Rectangle, Oval, or Freehand lines
c) Includes a bouncing ball animation using a Thread, controllable via buttons (Start / Stop)

Q28 Write a Java Swing Application to Create a Login Form:,design and implement a Swing-based login form that validates a username and password using event handling.

Q29 Write a java program to establish jdbc connection with MySql database and access the data from the database.

Q30. Write a java program to establish jdbc connection with MySql database and access the data from the database using JOIN query.

Q31. Write a step by step Spring Framework Application Concept :

- Create a simple Spring Core application using XML-based configuration that performs the following:
- Define a Java class named MessageService with a message property and a method showMessage() to print the message.
- Use Spring's XML configuration to inject a message like "Welcome to Spring Framework" into the MessageService bean.
- In the main class, load the Spring context from the XML file, retrieve the bean, and call the showMessage() method..

Q32. Write a step by step Java Bean Application concept by creating a JavaBean named Student with properties: name, rollNo, and course.

Q33. Write a Java Program to Create a Stateless Session Bean named SimpleCalculatorBean with methods to perform addition and subtraction.

Q34. Write a step by step Java Servlet Application to Create a servlet that displays a welcome message when a user accesses the URL.

Q35. Develop a Java Servlet Application to Create a servlet that maintains a counter of how many times it has been accessed.

Q36. Write a step by step Java Server Application to Create a JSP page that takes user input for name and email through an HTML form. On submission, display the submitted data on another JSP page using request.getParameter().

Q37. Write a step by step Java Server Application to Create a login form using JSP. When the user logs in, store their username in a session. Create a welcome page that displays "Welcome, [username]" using session.getAttribute().

Q38. Write a program to Create a JSP page that throws an exception intentionally (e.g., division by zero).

Create an error.jsp page to handle the exception gracefully using isErrorPage="true" and <%@ page errorPage="error.jsp" %> directive.

Q39. Write a step by step Java Network Application to know IP Address,port number,local ip address,local port number etc(system should be connected to Internet).

Q40. Q. Write Java programs for a TCP server and TCP client where the client sends the message “Hello Server!” and the server prints it. Use ServerSocket, Socket, DataInputStream, and DataOutputStream.

Or

Q40 . Using Java networking, develop a simple TCP Server and TCP Client application. The server must:

- Create a ServerSocket on port 5000
- Accept a client connection
- Read a message sent by the client using DataInputStream
- Display the received message

The client must:

- Connect to localhost on port 5000
- Send the message “Hello Server!” using DataOutputStream